

Introduction to Set

CS1010 Discrete Mathematics for Computer Science

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Motivation: Mathematical Structures in Computer Science

Computer Science is built upon a few fundamental mathematical structures.

► Sets

- ▷ Collections of distinct objects.
- ▷ Used to represent states, languages, graphs, databases, events etc.

► Sequences

- ▷ Ordered collections of elements.
- ▷ Used to model strings, arrays, execution traces, algorithms etc.

► Matrices

- ▷ Another way to represent structured data.
- ▷ Used in machine learning, graphics, scientific computing, cryptography etc.

► Functions

- ▷ Describe relationships between mathematical objects.
- ▷ Every algorithm can be viewed as a function mapping inputs to outputs.



Set

What is a Set?

Definition

A **set** is an unordered collection of objects.

- ▶ **Elements:** The objects in a set are called the *elements* or *members*.
- ▶ **Notation:**
 - ▷ $a \in A$ denotes that a is an element of the set A .
 - ▷ $a \notin A$ denotes that a is not an element of A .

ex. The students in this class

ex. The integers less than 10

ex. The vowels in the English alphabet

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Describing a Set: Notations

- ▶ **The Roster Method:** List all elements, separated by commas inside braces.

$$S = \{a, b, c, d\}$$

- ▶ **Order is irrelevant:** $\{a, b, c, d\} = \{b, c, a, d\}$
- ▶ Elements are either members or not; repetition doesn't mean anything.

$$\{a, b, c, d\} = \{a, b, c, b, c, d\}$$

- ▶ **Ellipses (...):** Used to describe massive or infinite collections when the pattern is completely clear.

$$S = \{0, 1, \dots, 100\}$$

Roster Method: Examples

ex. Prime numbers less than 15: $P = \{2, 3, 5, 7, 11, 13\}$

ex. Perfect squares less than 200: $K = \{1, 4, 9, 16, \dots, 169, 196\}$

ex. Multiples of 5 greater than 0: $F = \{5, 10, 15, 20, \dots\}$

Some Important Mathematical Sets

$\mathbb{N}$ Natural numbers = $\{0, 1, 2, 3, \dots\}$

$\mathbb{Z}$ Integers = $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

$\mathbb{Z}^+$ Positive integers = $\{1, 2, 3, \dots\}$

$\mathbb{Q}$ Rational numbers = $\{p/q \mid p, q \in \mathbb{Z}, q \neq 0\}$

$\mathbb{R}$ Real numbers

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Set-Builder Notation

Set-Builder notation specifies the properties that all elements of the set must satisfy, using a vertical bar '|' or colon ':' which translates to "such that". So this says: **the set of all x such that x satisfies this property.**

ex. $E = \{x \mid x \text{ is an even integer less than } 10\} = \{0, 2, 4, 6, 8\}$

ex. **Neg rationals:** $\mathbb{Q}^- = \{x \in \mathbb{R} \mid x = -p/q \text{ for some positive integers } p, q\}$

ex. $F = \{x \mid x \text{ is a perfect square less than } 50\} = \{0, 1, 4, 9, 16, 25, 36, 49\}$

ex. More formally: $F = \{x \in \mathbb{Z}^+ \mid \exists k \in \mathbb{Z} (x = k^2) \text{ and } x < 50\}$

ex. Predicate format: $S = \{x \mid P(x)\}$

ex. **Example:** $S = \{x \mid \text{PrimeInteger}(x)\}$

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Key Idea

The **roster method** tells us *which elements* are in a set, whereas **set-builder notation** tells us *what property* an element must satisfy to belong to the set.

Interval Notation

For real numbers, intervals are denoted with brackets or parentheses. A **bracket** means “include the endpoint,” while a **parenthesis** means “exclude the endpoint.”

- **Closed Interval** $[a, b]$: Includes both endpoints.

$$[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$$

- **Open Interval** (a, b) : Excludes both endpoints.

$$(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$$

- **Half-Open Interval** $[a, b)$:

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- **Half-Open Interval** $(a, b]$:

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Quick Check: Sets and Notation

Q. Which of the following is a set?

$\{1, 2, 3, 4\}$ “the students who are happy today”

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$\{2, 4, 6, 8, 10\}.$

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A. $\{x \in \mathbb{Z} \mid x \text{ is even and } 2 \leq x \leq 10\}.$



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A. $(3, 8]$

Sets: Structure and Visualization

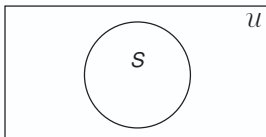
Universal Set, Empty Set, and Venn Diagrams

- ▶ **Universal Set (U):** Depending on the context, it is the set containing all elements of interest.
- ▶ **Empty Set ($\emptyset$):** The unique set containing no elements.

Venn Diagrams

Sets are visually represented inside a rectangle (universal set U) containing circles denoting subsets.

Example: $U = \mathbb{N}$ and $S = \{x \mid 1 < x < 10\}$.



Important Structural Rules to Remember

► Sets can be elements of other sets:

- ▷ $S = \{\{1, 2, 3\}, a, \{b, c\}\}$
- ▷ $\mathcal{N} = \{\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}\}$

► The Empty Set vs. the Set Containing the Empty Set:

- ▷ $\emptyset$ is the empty set (contains 0 elements).
- ▷ $\{\emptyset\}$ is a set containing one element (which happens to be the empty set).
- ▷ Therefore, $\emptyset \neq \{\emptyset\}$.

Russell's Paradox of Self Reference

Naïve set-builder without restriction can lead to mathematical contradictions.

The Paradox

Let S be the set of all sets that are not elements of themselves:

$$S = \{A \mid A \notin A\}$$

Question: Is S an element of itself? (Is $S \in S$?)

- ▶ If $S \in S$, then by definition, $S \notin S$ (Contradiction).
- ▶ If $S \notin S$, then by definition, $S \in S$ (Contradiction).

The Barber Analogy: A barber shaves everyone in town who does not shave themselves. Does the barber shave himself?



Axiomatic Set Theory

Russell's paradox shows that we cannot simply declare

$$S = \{x \mid P(x)\}$$

to be a set for every arbitrary property $P(x)$. We need precise rules governing which collections are allowed to be sets.

Sets: collections which can be proved from the axioms mentioned. One of the most widely used foundations is **Zermelo-Fraenkel set theory with the Axiom of Choice**, usually abbreviated as **ZFC**.

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Set Equality

We have seen that **order** and **repetition** do not matter. This naturally leads to the question: *When are two sets equal?*

Definition

Two sets A and B are **equal** if and only if they contain the exact same elements: $\forall x (x \in A \leftrightarrow x \in B)$. We write $A = B$ if they are equal sets.

ex. $\{1, 3, 5\} = \{3, 5, 1\}$

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The main question: Does every element of A belong to B , and does every element of B belong to A



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Subsets

Definition

The set A is a **subset** of B , denoted by $A \subseteq B$, if and only if every element of A is also an element of B : $A \subseteq B \iff \forall x(x \in A \rightarrow x \in B)$

- **Empty Set Rule:** Because the antecedent $a \in \emptyset$ is always false, the implication $a \in \emptyset \rightarrow a \in S$ is vacuously true. Hence,

$$\emptyset \subseteq S \quad \text{for every set } S$$

- **Self-Subset Rule:** Because $a \in S \rightarrow a \in S$ is always true:

$$S \subseteq S \quad \text{for every set } S$$

Proving Subsets (and Disproving)

- **To show that $A \subseteq B$:** Show that if $x \in A$, then x must also belong to B .
- **To show that $A \not\subseteq B$:** By counterexample: find x such that $x \in A$ but $x \notin B$.

ex. Let $A = \{2, 4, 6\}$ and $B = \{1, 2, 3, 4, 5, 6\}$ then $A \subseteq B$

ex. Let $A = \{1, 2, 3\}$ and $B = \{1, 2, 4, 5\}$. Since $3 \in A$ but $3 \notin B$, $A \not\subseteq B$.

Double Inclusion Proof Strategy

To prove $A = B$, you must show:

$$A \subseteq B \quad \text{and} \quad B \subseteq A.$$

Which means: $\forall x [(x \in A \rightarrow x \in B) \wedge (x \in B \rightarrow x \in A)]$

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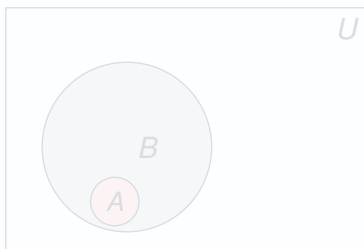
Proper Subsets

Definition

If $A \subseteq B$ but $A \neq B$, we say A is a **proper subset** of B , denoted by $A \subset B$.

Formally: $A \subset B$ if and only if $\forall x, (x \in A \rightarrow x \in B) \wedge \exists x(x \in B \wedge x \notin A)$

ex. Consider $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4\}$ then $A \subset B$



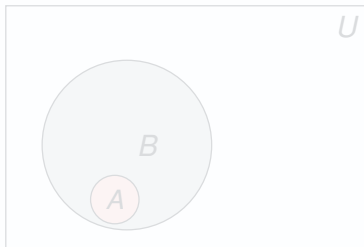
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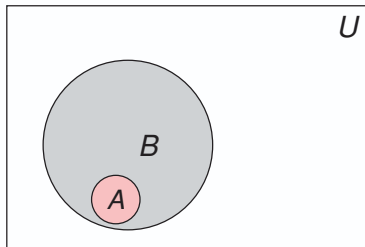
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Set Cardinality

Cardinality

The cardinality of a finite set S , denoted by $|S|$, is the number of distinct elements in S .

ex. $|\emptyset| = 0$

ex. Let S be the letters of the English alphabet. $|S| = 26$.

ex. $|\{1, 2, 3\}| = 3$

ex. $|\{\emptyset\}| = 1$

ex. Not every set is finite. For example, the set of all integers

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

has infinitely many elements. We therefore say that $\mathbb{Z}$ is an **infinite set**.



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Power Sets

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Given a set A , the **power set** of A , denoted by $\mathcal{P}(A)$, is the set of all subsets of A .

ex. Let $A = \{a, b\}$. The subsets are $\emptyset, \{a\}, \{b\}, \{a, b\}$.

$$\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$$

Q. How many subsets are there?

Q. Prove that if A is a finite set with $|A| = n$, then $|\mathcal{P}(A)| = 2^n$.

A. Consider an element a . For any subset A , either $a \in A$ or $a \notin A$:

$$\underbrace{2 \times 2 \times \cdots \times 2}_{n \text{ times}} = 2^n$$

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A. Yes. The empty set is a subset of every set.

Quick Check: Equality and Subsets

Q. Are the following sets equal: $A = \{1, 2, 3\}$, and $B = \{3, 1, 2, 2, 1\}$?

A. Yes. Both sets contain exactly the elements 1, 2, 3.

Q. Let $A = \{1, 2\}$ and $B = \{1, 2, 3, 4\}$. Is $A \subseteq B$?

A. Yes. Every element of A is also an element of B .

Q. Is $\{1, 2, 3\} \subseteq \{1, 2, 3\}$?

A. Yes. Every set is a subset of itself.

Q. Is $\emptyset \subseteq A$ for every set A ?

A. Yes. The empty set is a subset of every set.

Q. Is $\emptyset = \{\emptyset\}$?

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Q. Is $\emptyset = \{\emptyset\}$?

A. No. $|\emptyset| = 0$, but $|\{\emptyset\}| = 1$.

Quick Check: Proving Subsets

Q. Let $A = \{2, 4, 6\}$, and $B = \{1, 2, 3, 4, 5, 6\}$. Is $A \subseteq B$?

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Q. Let $A = \{2, 4, 6\}$, and $B = \{1, 2, 3, 4, 5, 6\}$. Is $A \subseteq B$?

A. Yes. $2, 4, 6 \in B$, so every element of A belongs to B .

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A. Yes. $2, 4, 6 \in B$, so every element of A belongs to B .

Q. Let $A = \{1, 3, 5\}$, and $B = \{1, 2, 3, 4\}$. Is $A \subseteq B$?

A. No. For example, $5 \in A$ but $5 \notin B$. Thus, 5 is a counterexample.

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Q. Let $A = \{1, 2, \{1, 2\}\}$. Is $\{1, 2\} \in A$?

A. Yes.

Q. Is $\{1, 2\} \subseteq A$?

A. Yes. Both 1 and 2 are elements of A .

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Q. Let A and B be finite sets with $|A| = 5$, $|B| = 3$, and $A \subseteq B$. Is this possible?

A. No. If $A \subseteq B$, then A cannot have more elements than B . Hence,

$$A \subseteq B \implies |A| \leq |B|.$$

Quick Check: Power Sets

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A. $\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{\{1, 2\}\}, \{1, 2\}, \{1, \{1, 2\}\}, \{2, \{1, 2\}\}, \{1, 2, \{1, 2\}\}\}$.

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A. There are $2^4 = 16$ such subsets. Once a is required to be included, each of the remaining 4 elements can independently be included or excluded.

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A. We choose 2 elements from 4, i.e., 6 subsets.

Tuples

Ordered n -tuples

The **ordered n -tuple** $(a_1, a_2, \dots, a_n)$ is the ordered collection that has a_1 as its first element, a_2 as its second element, and so on up to a_n .

- ▶ Two n -tuples are equal if and only if their corresponding elements are equal.
- ▶ 2-tuples are called **ordered pairs**.
- ▶ $(a, b) = (c, d) \iff a = c \wedge b = d$.
- ▶ Note that while $\{a, b\} = \{b, a\}$, the ordered pairs $(a, b) \neq (b, a)$ unless $a = b$.

ex. A student could be represented as

(Name, Roll Number, Department, Year),

where each position has a specific meaning. Interchanging two positions would change the meaning of the entire tuple.



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Problem: The Crazy Passenger Puzzle

A line of n airline passengers is waiting to board a plane. The i -th passenger in line has a ticket for seat number i .

The Twist

- ▶ **Passenger 1:** Ignores his ticket, and chooses a completely random seat out of the n available seats.
- ▶ **Passengers 2 through n behave logically:** they will walk to their assigned seat. If it is free, they sit in it. If it is already occupied, they pick a remaining free seat completely at random.

Question: Write down the set of all possible final sitting arrangements as an explicit set notation.



Cartesian Product

Definition

The **Cartesian product** of sets A and B , denoted by $A \times B$, is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$.

$$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$$

Example: Let $A = \{a, b\}$ and $B = \{1, 2, 3\}$.

$$A \times B = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}$$

Example: Possible Realization of 3 tosses: $\{H, T\} \times \{H, T\} \times \{H, T\}$

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Quick Check: Cartesian Products

Q. Let $A = \{1, 2\}$, $B = \{x, y, z\}$. How many elements does $A \times B$ have?

Quick Check: Cartesian Products

Q. Let $A = \{1, 2\}$, $B = \{x, y, z\}$. How many elements does $A \times B$ have?

A. $|A \times B| = |A||B| = 2 \cdot 3 = 6$.

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Q. Let $A = \{1, 2\}$. How many elements does $A \times A$ have? List them.

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Q. Let $A = \{1, 2\}$. How many elements does $A \times A$ have? List them.

A. $A \times A = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$, so $|A \times A| = 4$.

Quick Check: Cartesian Products

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Q. Let $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$. Is $(2, 3) \in A \times B$? Is $(3, 2) \in A \times B$?

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Q. Let $A = \{1, 2\}$, $B = \{x, y, z\}$. How many elements does $A \times B$ have?

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Q. Let $A = \{1, 2\}$. How many elements does $A \times A$ have? List them.

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Q. Let $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$. Is $(2, 3) \in A \times B$? Is $(3, 2) \in A \times B$?

A. Yes to both. We need the first component to be in A and the second component to be in B :

$$2, 3 \in A, \quad 3, 2 \in B.$$





Thank You for your
attention!

Any questions?